ALIoT Smart Lab - engineering dossier

Version 2026-10-20 (replaces nothing - the 2026-10-18 dossier stays as history). Branch `offline-router-door-photos`, nothing committed or pushed. Deployed portal version `f5eaeff-dirty-20261020004125` (from `/api/health`). Database migration head `b7c3e1f9a204`. Labels: LIVE-VERIFIED seen working on the real system, TESTED OFFLINE automated tests only, SIMULATED, PHYSICAL VALIDATION REQUIRED, BLOCKED / OWNER INPUT.

State while this was written: the laptop was off the lab network (phone hotspot). Both camera services run the new code but have no stream; no ESP node, robot or camera could be tested live after 19:15 on 2026-10-19. Portal changes were deployed and verified on the live portal; tracking changes are tested offline only.

1. What changed in this round

Area	Root cause (evidence)	Fix	Status
USER1 / USER5 cross-camera conflict	Camera 1's desk rule named whoever sat at PC-09/PC-10 "USER5" from a door entry hours old; camera 2's face said USER1. 4 episodes (16:14, 16:15, 17:02, 18:36), 243 conflict lines, 546 of 547 USER5 namings came from the desk.	The desk only continues evidence: name on a track within 120 s, or door / face / other camera within 10 min. Otherwise `DESK_NOT_USED`.	TESTED OFFLINE regression test fails on the old code
Seated-person flicker	The labeler saw only tracks detected in the last 0.6 s while the tracker keeps a still track 6 s: every short miss logged NAME_LOST and let the desk name a second track (NAME_ON_TWO_PEOPLE x24).	Tracks still kept by the tracker ("coasting") keep their names and are not reported lost; hand-back counts from last-seen time.	TESTED OFFLINE
Nour Unknown after a short track break	Hand-back looked only at the spot where the track was lost; a walking person reappears further along.	Search along the walking path (look must still agree).	LIVE-VERIFIED 19:05 walk test
Wrong name after a crossing (camera 2, 19:05)	Nour's name, correctly handed over, stayed on camera 2's track after it went onto Ali; camera 2's own face said USER1; by design the name was not switched.	A name only carried over (other camera, desk) gives way to this camera's 2 agreeing face matches when nobody touches. Door names keep the mismatch flag. Face thresholds unchanged.	TESTED OFFLINE
FANUC Educational Cell	Drawn from overall dimensions only; panels 12 % opaque; yellow panel on the wrong face; no lettering; cell front opposite the owner's real front.	Glass with edges, two doors, hinges, knob, FANUC sticker, yellow side panels ("Learn online: fanuc.eu/educational-package"), controller with e-stop, pendant, crate, slotted table, parts plate, "FANUC Robot ER-4iA" on the forearm; robot right of the plate, front = owner's front.	LIVE-VERIFIED deployed, screenshots

Robot mesh size	UR5e meshes were unindexed (each position stored ~5.6 times); the 2026-10-19 "12.5 MB -> 75 KB" counted only the index files.	Weld + quantize, every triangle kept: UR5e + Robotiq 10.4 -> 1.9 MB, FANUC 1.26 -> 0.26 MB.	LIVE-VERIFIED renders identical
System tab slow	12 Recharts charts built at once (643 ms block) plus entry animations (~100 ms blocks repeating).	Charts built when near the screen; animations off on that page.	LIVE-VERIFIED
Evacuation alarm missing on full-screen pages	Alarm and emergency message mounted only in the portal frame - /welcome (every sign-in lands there), /monitor, /wall, /report showed neither.	Safety layer on those pages; Welcome gets live updates.	TESTED OFFLINE e2e on the isolated copy; deployed
Stale data after a write	Identical GETs shared one answer for 1 s, also across a write: a photo uploaded within 1 s of opening an issue stayed invisible.	Every write clears the shared reads.	TESTED OFFLINE workflows 30/30; deployed
Welcome finale crash (own regression)	The mesh-loader change sent Kinova `.STL` files through the glTF loader (deployed 23:36).	Only the converted FANUC uses .glb; fixed and redeployed 00:01.	LIVE-VERIFIED welcome 20/20 admin+staff

2. Measurements

Portal navigation (deployed build, real browser, admin)

Page	Before (first / repeat)	After (first / repeat)
Robots & PCs tab (cold, empty cache)	2.0 s, 8.3 MB models	1.5 s, ~1 MB models
Robots & PCs tab (repeat visit)	0.78 s	0.29 s
System analytics	1.20 / 0.88 s	0.27 / 0.23 s
Reliability	0.96 / 0.84 s	0.08 / 0.04 s
All other sidebar tabs	66-693 ms	40-500 ms

Medians of 3 runs (robots) / 2 rounds (tabs). Scripts: `frontend/e2e/robots_tab_perf.mjs`, `nav_timing.mjs`. Remaining cold cost is main-thread work (3D library start, WebGL), not downloads.

Load test SIMULATED (isolated backend + test DB, 120 s)

Measure	Result
API errors	0
API p95	all < 220 ms; `/summary` one sample of 12 at 525 ms; `/monitor` p95 215 ms
Two simulated cameras (GPU, DirectML)	6.5 / 6.9 analyses/s, frame age p95 ~85 ms, GPU 50 %, VRAM 1155 MB
Memory / threads	RSS 2163 -> 2171 MB, threads 300 -> 283 (no growth)
WebSocket per client	robot_status 1.1/s, env 0.36/s, security 0.19/s

AI assistants LIVE-VERIFIED

Structured answers 0.0-0.2 s; document lookup 1.6 s; an answer from the language model (qwen2.5:1.5b on the **CPU**, cameras idle) 51.5 s. CPU is the owner's choice (GPU for the cameras; GPU answers were ~3 s on 2026-10-19).

Cross-camera evidence (camera logs 2026-10-19)

Measure	Value
Appearance distance, same person (both names confirmed)	n = 119, median 0.30, p90 0.44, max 0.73
Appearance distance, different people	n = 2 (0.67, 0.70) - the earlier 0.28 / 0.38 pairs used the wrong desk name and are excluded
Floor gap, same person in both cameras	median 0.41 m, p90 0.86 m (pairing gate 0.7 m)
Floor coverage from the two floor maps	both cameras: middle of the room (y ~3.5-8 m); camera 1 only: far end; camera 2 only: near end; neither ~1 % (occlusion not modelled)
Handover at entry (19:04 walk test)	camera 1 named at the door, camera 2 had the name 4 s later; longest Unknown on camera 2: 35 s after a track break where only camera 2 saw the person
Simulated scenario bench (14 scenarios)	no false names, no name switches - identical to 2026-10-18 SIMULATED

Conclusion: appearance matching stays observe-only - not because it is shown unreliable, but because only 2 different-person pairs exist. The blind-spot Unknown time is track breaks in camera 2's own area, not unseen floor.

3. Tests

Suite	Result
Backend (pytest, smartlab_test)	542 passed (8 min 19 s)
Security cameras	253 passed (11 new regression tests)
Robot monitor / knowledge / PC agent	41 / 40 / 5 passed
Frontend	TypeScript clean; production build OK
Browser e2e, isolated seeded copy	workflows 30/30, welcome 27/27, reliability 31/31, monitor+security 18/18, emergency message 6/6, notifications, experiments, report link pass; navigation 97 checks, 2 console messages = camera 503 (test cameras stopped)
Browser e2e, live portal (read-only)	robot readiness 8/8, monitor+security 18/18, reliability 31/31, welcome 20/20 (admin, staff)
Not run	camera recording e2e (cameras offline); student-role checks on live (real account, not the demo password)

Six e2e scripts were updated for UI changes made since they were written (sign-in now lands on /welcome; booking wizard has an Equipment step; lab count read from the database).

4. Devices

Device	State
Door master, environment node, consumables bin, reusable bin, PC agent	last report 2026-10-19 03:59-04:29 local; **none reported during the 2026-10-19 lab session** although the cameras on the same network did PHYSICAL CHECK
ESP32-CAM + face server	face server up (SFace, threshold 0.363); templates USER1, 2, 3, 5, 6, 7 - **Hemaly (USER5) is loaded** (23 samples); live door test needs her present PHYSICAL
Robots (UR5e, FANUC)	collectors running read-only, no data off the lab network; DRY RUN, nothing sent
Cameras	services healthy, reconnecting; 7 fps each when live (2026-10-19)

5. Remaining - owner actions

#	Item	Type
1	Walk tests at the lab (one person, then two crossing) to confirm the tracking fixes live; record a 2-person session for appearance thresholds	PHYSICAL
2	Only if the walk shows long camera-2-only stretches: tilt camera 2 and re-click the tape marks (back up the floor JSON first)	PHYSICAL
3	PC letter -> number mapping for camera 2 (B, C, D, F, G, I, J, K, L -> PC-01..11)	OWNER INPUT
4	Hemaly at the door: positive + negative face test	PHYSICAL
5	Why the ESP nodes stopped reporting at 04:00 (power, Wi-Fi, portal address)	PHYSICAL
6	Change the live admin password - it equals the demo seed password published in the public repo	OWNER
7	Door wiring values not recorded (section 6) - measure	PHYSICAL
8	Commit when the owner approves	OWNER

6. Appendix - door master (MASTER_LAB01) wiring as recorded

From the firmware constants and the 2026-10-04/05 bring-up log; full version with parts table: `docs/hardware/DOOR_MASTER_WIRING.md`. "Not recorded" = measure before use; nothing was guessed.

ESP32 GPIO	Connected to	Notes
19 / 18	PN532 V3 (HW-147) SDA / SCL	I2C 0x24, S1 ON / S2 OFF, IRQ + RESET not wired; supply pin not recorded
13 / 2	ST7735 1.8" TFT SCK / MOSI	own SPI bus; GPIO2 is a boot strapping pin
25 / 32 / 33	ST7735 CS / DC / RST	VCC and backlight supply not recorded
16 / 17	fingerprint TX -> ESP RX16; ESP TX17 -> sensor RX	AS608-protocol, 57600 8N1; model and supply not recorded
27	MC-38 reed switch to GND	INPUT_PULLUP: LOW closed, HIGH open

26	BC557 base via 3.3 kΩ	HIGH = locked, LOW = unlocked
14	green LED	resistor not recorded
21	NPN stage: red LED + buzzer	transistor and resistors not recorded; 600 ms denial beep
15	red LED via 220 Ω to GND	own pin, no buzzer current
22	3.3 kΩ series -> PAM8403 L-in, 100 nF to GND	3 W / 8 Ω speaker on L OUT+/-; log says removed 2026-10-05 - confirm

```
+12 V -- relay VCC, relay COM, 470 uF 25 V (+) ;  (-) -- GND
relay NO -- solenoid (+) ;  solenoid (-) -- GND            (lock on NO = fail-secure)
3V3 -- BC557 E ;  BC557 C -- relay IN ;  BC557 B -- 3.3 k -- GPIO26 ;  15 k between B and E
GPIO26 HIGH -> PNP off -> coil off -> locked.   LOW -> ~1 mA base -> coil on -> unlocked.
Measured: COM-GND 12.1 V at rest, 12.03 -> 11.95 V during an unlock.
Supply: 12 V / 10 A PSU -> master switch -> WAGO -> fused branch -> buck -> 5 V -> ESP32 VIN (door
buck model not recorded).
```

Records disagree: `door_audio.h` says RC ~47 µs, but 3.3 kΩ x 100 nF = 330 µs; `docs/DOOR_SYSTEM.md` pin table still shows the retired MFRC522 and the lock on NC.

7. Evidence files

`docs/camera-identity/TWO_CAMERAS.md` (2026-10-20 section), `docs/PROGRESS_CHECKPOINT.md`, `docs/hardware/DOOR_MASTER_WIRING.md`, `tools/security_vision/test_desk_evidence_2026_10_19.py`, `test_face_overrides_carried_name.py`, `test_walk_handback.py`, `frontend/scripts/compact_glb.mjs`, `stl_to_glb.mjs`, `frontend/e2e/robots_tab_perf.mjs`, `backups/deploy_history.jsonl`. Camera logs (git-ignored): `tools/security_vision/recordings/identity_log_*.jsonl`, `crosscam_obs_*.jsonl`.